

Full Stack Development with MERN

Project Documentation

1. Introduction

Project Title: Stock Trading Application

Team Members

Name	Role
Parnapalli Sabiya Kousar	Team Leader
Mochi Harini	Team Member
Sake Yaswanth	Team Member
Kuruba Keerthi	Team Member
Rayankula Nikhitha	Team Member

2. Project Overview

Purpose

The **Stock Trading Application** is a MERN Stack web application developed to provide users with a secure and interactive platform for monitoring stock market data, buying and selling stocks, managing investment portfolios, and maintaining personalized watchlists. The project demonstrates modern full-stack web development using MongoDB, Express.js, React.js, and Node.js.

Features

- User Registration and Login
- JWT Authentication
- Dashboard
- View Stock List
- Search Stocks
- Interactive Stock Charts

- Buy Stocks
- Sell Stocks
- Portfolio Management
- Watchlist
- Transaction History
- Admin Panel
- Responsive User Interface

3. Architecture

Frontend

The frontend is developed using **React.js** with reusable components and React Router for navigation. The user interface is responsive and communicates with the backend through REST APIs.

Technologies Used

- React.js
- HTML5
- CSS3
- JavaScript
- Axios
- Chart.js

Backend

The backend is built using **Node.js** and **Express.js**. It provides RESTful APIs for authentication, stock management, portfolio management, and transactions.

Technologies Used

- Node.js
- Express.js
- JWT Authentication
- bcrypt.js

Database

MongoDB Atlas is used as the cloud database.

Collections include:

- Users
- Stocks
- Portfolio
- Transactions
- Watchlist

4. Setup Instructions

Prerequisites

- Node.js
- npm
- MongoDB Atlas Account
- Git
- Visual Studio Code

Installation

Clone Repository

```
git clone https://github.com/ParnapalliSabiyaKousar/Stock-Trading-App.git
```

Backend Setup

```
cd server
```

```
npm install
```

Create a **.env** file

```
PORT=5000
```

```
MONGO_URI=your_mongodb_connection_string
```

```
JWT_SECRET=your_secret_key
```

Run Backend

```
npm start
```

Frontend Setup

cd client

npm install

npm run dev

5. Folder Structure

Stock-Trading-App

client/

|

├── src/

|

├── components/

|

├── pages/

|

├── services/

|

├── styles/

|

├── App.jsx

|

└── main.jsx

|

server/

|

├── config/

├── controllers/

├── middleware/

├── models/

├── routes/

├── uploads/

└── server.js

└─ package.json

6. Running the Application

Backend

cd server

npm start

Runs on:

http://localhost:5000

Frontend

cd client

npm run dev

Runs on:

http://localhost:5173

7. API Documentation

Method	Endpoint	Description
POST	/api/auth/register	Register User
POST	/api/auth/login	User Login
GET	/api/stocks	Get All Stocks
GET	/api/stocks/:id	Get Stock Details
POST	/api/stocks/buy	Buy Stock
POST	/api/stocks/sell	Sell Stock
GET	/api/portfolio	View Portfolio
GET	/api/watchlist	View Watchlist
POST	/api/watchlist	Add to Watchlist
DELETE	/api/watchlist/:id	Remove Watchlist Item

Method	Endpoint	Description
POST	/api/admin/add-stock	Add Stock (Admin)
DELETE	/api/admin/delete-stock/:id	Delete Stock

8. Authentication

Authentication is implemented using **JWT (JSON Web Token)**.

Security Features:

- User Registration
- Password Encryption using bcrypt.js
- JWT Token Generation
- Protected Routes
- Admin Authorization
- Session Validation

9. User Interface

The application includes the following screens:

- Landing Page
- Registration Page
- Login Page
- Dashboard
- Stock Listing
- Stock Details
- Stock Charts
- Portfolio
- Watchlist
- Buy/Sell Page
- Admin Dashboard

10. Testing

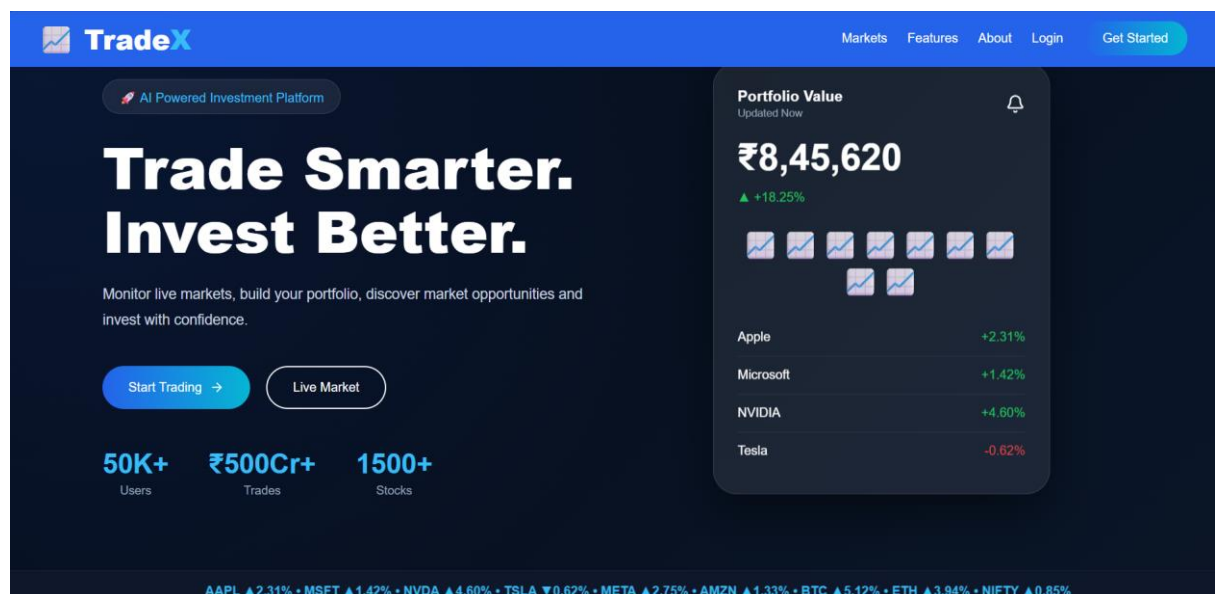
Testing was carried out using:

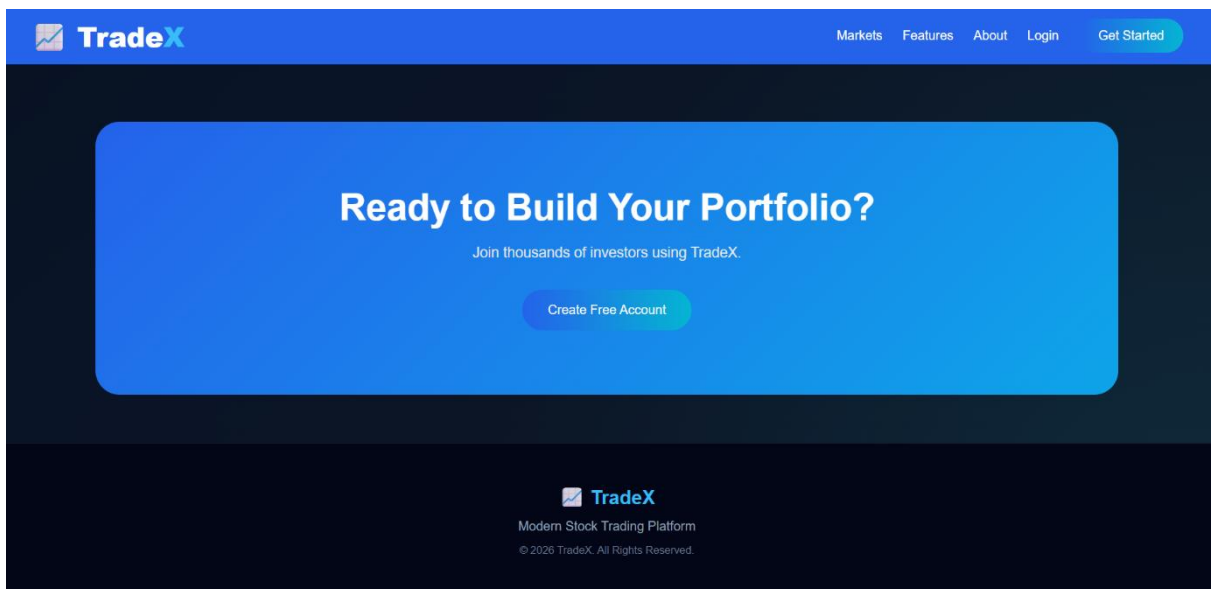
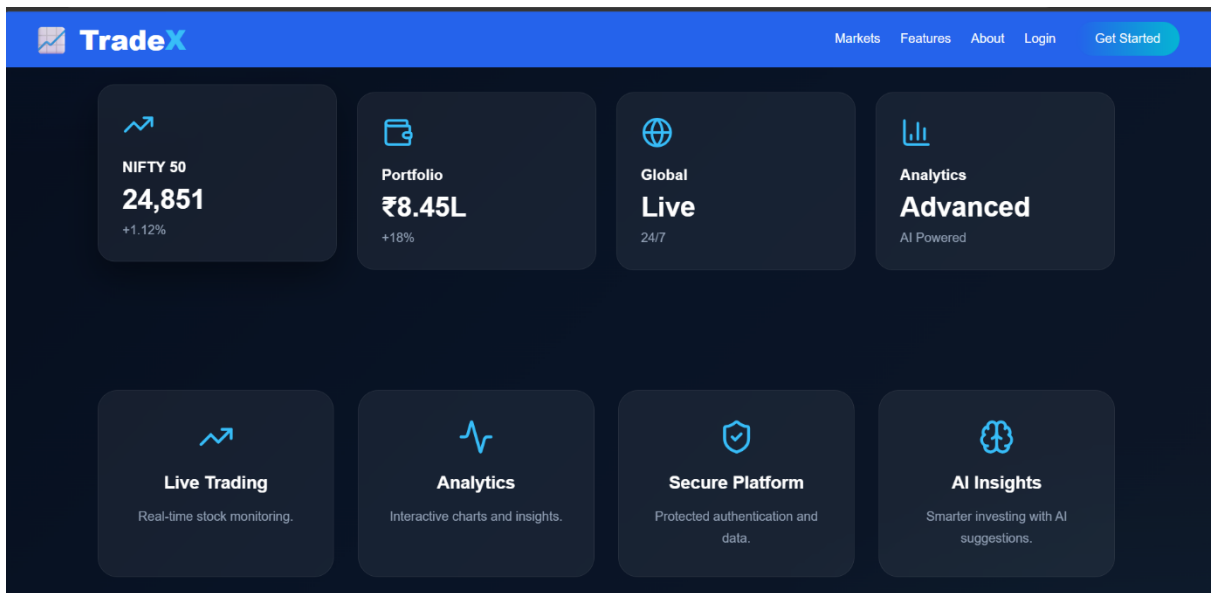
- Manual Functional Testing
- User Acceptance Testing (UAT)
- API Testing using Postman
- Frontend Testing in Chrome and Microsoft Edge
- Database Validation using MongoDB Atlas

11. Screenshots / Demo

Include screenshots of:

- Home Page
- Login
- Registration
- Dashboard
- Stock List
- Buy Stock
- Sell Stock
- Portfolio
- Watchlist
- Admin Panel





SB Stocks Create Account

Join and start investing today.

Full Name

Email Address

Password

Confirm Password

[Register](#)

Already have an account? [Login](#)



SB Stocks

Welcome Back

Login to continue trading.

Login

Don't have an account? [Register](#)

[Dashboard](#) [Markets](#) [Portfolio](#) [Watchlist](#) S Sabiya

Welcome Back

Monitor your investments in one place.

S Sabiya Investor

Portfolio Value
\$25,430.50
▲ +3.8%

Today's Profit
\$540.20
+\$120 Today

Stocks Owned
24
Across 6 sectors

Watchlist [View All](#)

Apple AAPL	\$215.36 +2.15%
Microsoft MSFT	\$512.18 +1.24%
Tesla TSLA	\$318.80 -0.64%
NVIDIA NVDA	\$174.82 +4.31%

[Dashboard](#) [Markets](#) [Portfolio](#) [Watchlist](#) S Sabiya

Stock Market

Apple
AAPL
\$215.36
+2.15%
[View Chart](#)

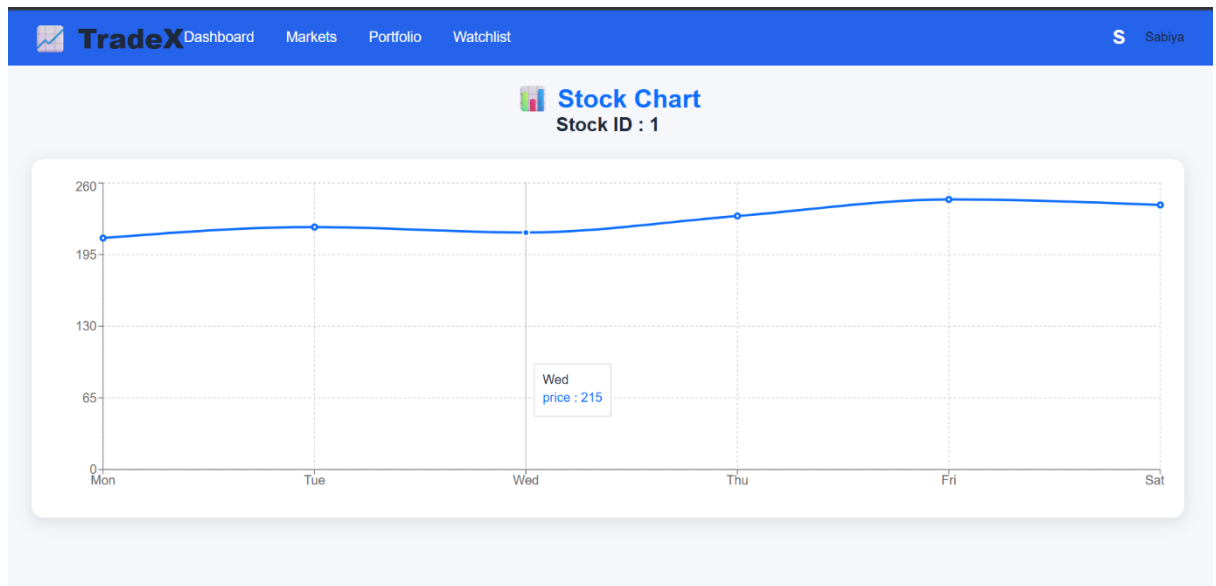
Microsoft
MSFT
\$512.18
+1.24%
[View Chart](#)

Tesla
TSLA
\$318.80
-0.64%
[View Chart](#)

Amazon
AMZN
\$228.14
+0.91%
[View Chart](#)

NVIDIA
NVDA
\$174.82
+4.31%
[View Chart](#)

Google
GOOGL
\$181.70
+1.55%
[View Chart](#)



GitHub Repository:

<https://github.com/ParnapalliSabiyaKousar/Stock-Trading-App>

12. Known Issues

- Live stock prices depend on API availability.
- Network latency may affect chart loading.
- Invalid API keys may prevent stock data retrieval.
- Admin access is restricted to authorized users.

13. Future Enhancements

Future improvements include:

- Integration of real-time stock market APIs.
- AI-based stock price prediction.
- Mobile application using React Native.
- Email and SMS notifications.
- Multi-language support.
- Payment gateway integration.
- Advanced portfolio analytics.
- Dark and Light Theme support.

- Two-Factor Authentication (2FA).
 - Cloud deployment with Docker and Kubernetes.
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Conclusion

The **Stock Trading Application** successfully demonstrates the implementation of a secure and scalable MERN Stack web application. It provides essential stock trading functionalities, including authentication, stock management, portfolio tracking, watchlist management, and an administrative interface. The project follows modern software engineering practices and can be extended with advanced features in future releases.